

Phoenix and the ***WORLD OF TOMORROW***



THE NEXT BILLION

More than 3 Billion Smart Phones active right now

Many of the users are coming online for the first time

The infrastructure is present but could be quite slow

The rate of growth is staggering

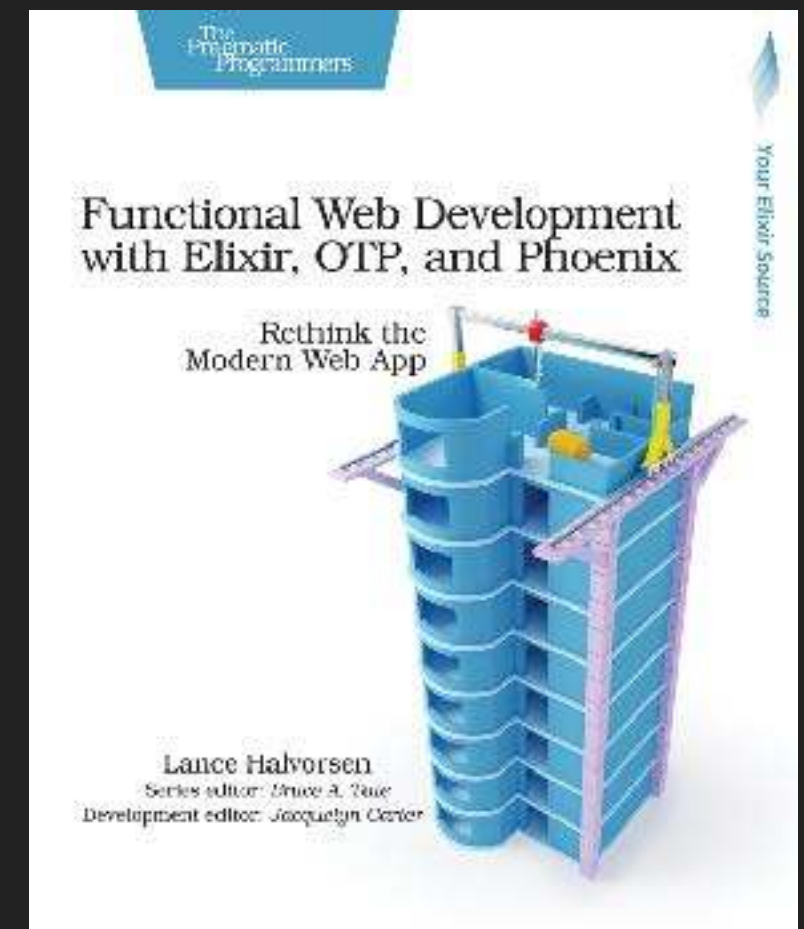
The price of bandwidth is high in both price and time

PHOENIX IS NOT YOUR APPLICATION

Phoenix is the web interface solution



Lance Halvorsen - Phoenix Is Not Your Application (ElixirConfEU 2016)



THE EDGE OF YOUR APPLICATION IS NOT THE DATA CENTRE

Phoenix is fast to getting data off our server

... but we don't think about when in lands.

When the user gets to interact with you is the important bit.

THE EDGE OF YOUR APPLICATION IS NOT THE DATA CENTRE

Speedcurve can help with that.



MOBILE FIRST DESIGN

Go out and buy a low end Android phone

Google recommends Moto G4

Always keep the Bandwidth throttled to 2G Max

BUDGETS

Two forms of budgets other than monetary

Time

Footprint

BUDGETS

Units of



2.39MB

BUDGETS

Time

Not all bits are created equal

EVERYTHING on your page must do a job

Two Goal Posts

Page load

Full appreciated experience

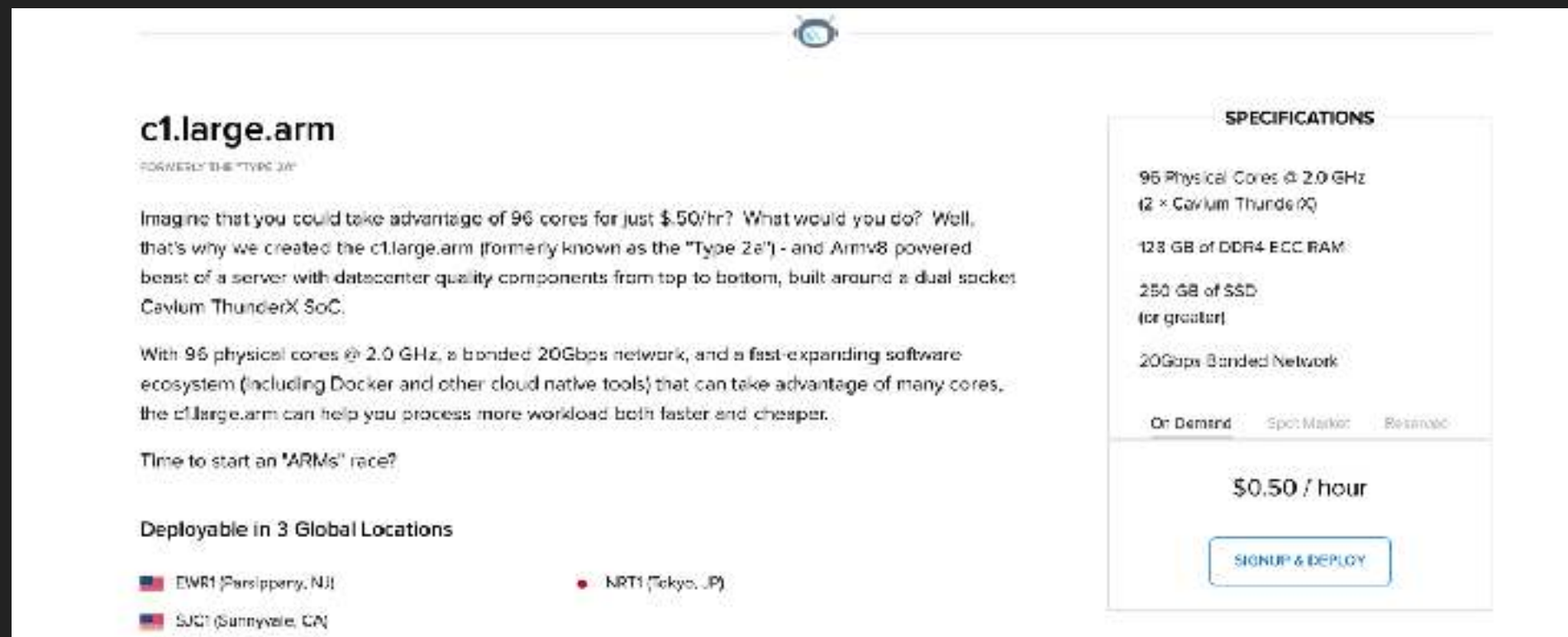
IN THE BEGINNING

There was that server and the sysadmin saw that is was good. And then the developer said let there be apps ...

BARE METAL AND NEIGHBOURS

Dramatic speed boost to actually be on hardware

No noisy neighbours



The screenshot shows the AWS console page for the c1.large.arm instance type. The page is titled 'c1.large.arm' with a sub-header 'FORMERLY THE "TYPE 3A"'. The main text describes the instance as a server with datacenter quality components, built around a dual socket Cavium ThunderX SoC. It highlights 96 physical cores @ 2.0 GHz, 128 GB of DDR4 ECC RAM, 250 GB of SSD (or greater), and a 20Gbps bonded network. The pricing is shown as \$0.50 / hour, with a 'SIGNUP & DEPLOY' button. The page also lists three global locations: EWR1 (Parsippany, NJ), SJC1 (Sunnyvale, CA), and NRT1 (Tokyo, JP).

c1.large.arm
FORMERLY THE "TYPE 3A"

Imagine that you could take advantage of 96 cores for just \$.50/hr? What would you do? Well, that's why we created the c1.large.arm (formerly known as the "Type 2a") - and Armv8 powered beast of a server with datacenter quality components from top to bottom, built around a dual socket Cavium ThunderX SoC.

With 96 physical cores @ 2.0 GHz, a bonded 20Gbps network, and a fast-expanding software ecosystem (including Docker and other cloud native tools) that can take advantage of many cores, the c1.large.arm can help you process more workload both faster and cheaper.

Time to start an "ARMs" race?

Deployable in 3 Global Locations

- 🇺🇸 EWR1 (Parsippany, NJ)
- 🇯🇵 NRT1 (Tokyo, JP)
- 🇺🇸 SJC1 (Sunnyvale, CA)

SPECIFICATIONS

- 96 Physical Cores @ 2.0 GHz (2 x Cavium ThunderX)
- 128 GB of DDR4 ECC RAM
- 250 GB of SSD (or greater)
- 20Gbps Bonded Network

On Demand | Spot Market | Reserved

\$0.50 / hour

[SIGNUP & DEPLOY](#)

FREEBSD AND ZFS

Just do it

HAPROXY ET AL.

Use a proxy outside the VM

COMPRESSION

Netflix saved 43% of their bandwidth by turning on gzip compression. :O

GZIP AND BROTLI

Gzip is Ubiquitous

So is Brotli

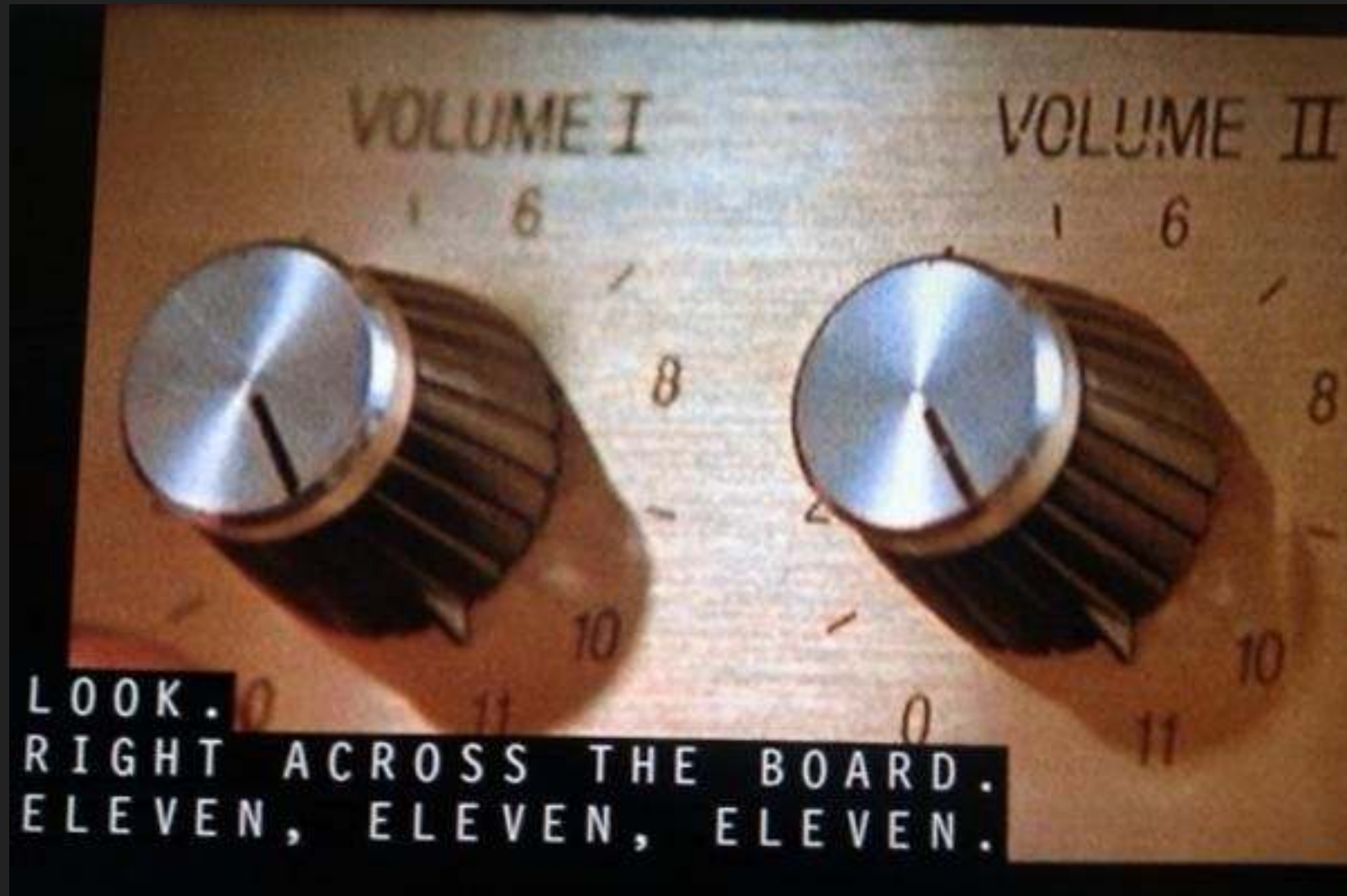
However ...

- HTTPS only

- Can be slower

- But not really

DIAL IT TO 11



HTTPS

Harfleur on Hex Right Now

```
def call(conn, _opts \\ []) do
  breach = :rand.uniform(128)
  |> :crypto.strong_rand_bytes()
  |> :base64.encode()

  conn
  |> Conn.put_resp_header("x-welcome-robot-overlords", breach)
end
```


WEBSOCKETS - THERE BE DRAGONS

```
Once upon a midnight dreary, while I {pon}dered weak an{d wea}{ry,}
Over many{ a }quaint{ and }curious volume of forgotten lore,
W{hile I }nodded, n{ear}ly napping, su{dde}n{ly }th{ere} ca{me }a t{apping,}
As{ of }so{me o}ne gent{ly }r{apping, }{rapping} at my chamb{er }door.
`Tis{ some }visitor,`{ I }mu{tte}r{ed, }`t{apping at my chamber door} -
O{nly th}is,{ and }no{thi}{ng }m{ore}.'
```

```
Ah, distinc{tly }I reme{mber }it was in{ the }bl{eak }Dec{ember}{,
A}{nd }each separate dy{ing }{ember }wroug{ht }its ghost{ upon }{the }fl{oor.
}Eager{ly I }wish{ed }{the }{mor}row; - v{ain}{ly I }had{ so}{ught }to b{orrow}
From{ my }book{s s}urcease{ of so}{rrow}{ - }{sorrow }{for}{ the }l{ost }Len{ore}{ -
}F{or the }ra{re }{and }radia{nt }mai{den}{ wh}{om }{the }angels{ na}{me }{Lenore -
}N{ame}less {here }{for }e{ver}{more.}
{
And }{the }silk{en }s{ad }u{n}ce{rt{ain} rustl{ing }{of }{each }purp{le }{cur}{tain}
Thrill{ed }{me }- f{illed me }with f{ant}a{sti}c {ter}ror{s n}{ever} felt be{for}e;
So{ th}{at }now,{ to }{sti}ll{ the b}eat{ing of }{my }h{ear}t,{ I }stoo{d r}ep{eating}{
`Tis some visitor} {ent}r{eating }{entr}a{n}ce }{at my chamber door -
}S{ome }l{ate }{visitor entreating entrance at my chamber door}{; -}{
Th}{is }{it }{is, and nothing more},{

```

```
}Pres{ently }{my }{sou}l grew{ st}ro{n}ge{r; }he{sit}{ating }{the}n{ no}{ lo}{nger},
`Si{r,` }s{aid} I{, `}{or }Madam{, t}ru{ly }your{ forg}iven{ess }I impl{ore;
}But{ the f}ac{t is}{ I w}as{ napping, }{and }so{ gently }{you}{ came }r{apping,
A}{nd so }f{aint}{ly you came }{tapping,} {tapping at my chamber door},{
Th}{at }{I s}ca{rce}{ was }{sur}{e I }{hear}d{ you}'{ - }{here }I op{ene}{d w}{ide}{ the }{door; -
}Dark{ness }{there}{, and nothing more.}
```


WAIT ISN'T THIS A PHOENIX TALK



PHOENIX TEMPLATING

EEX is both very fast and very good - USE IT!

Care about HTML Semantics - it's important for accessibility

Gettext - Same.

SOON TM

CSS - WHAT IS IT EVEN?

CSS
IS
AWESOME



CSS - HOW IT ACTUALLY WORKS

```
defmodule Browsercss do
  defstruct{ important: 0, inline: 0, id: 0, class: 0, elements: 0}

  def render(html, css, _javascript) do
    dom(html)
    |> cssom(css)
    |> layout()
    |> paint()
  end
end
```

CSS - CSS CLASS ONLY DESIGN

Inverted Pyramid

BEM naming structure

Boils CSS down

Harry Roberts - CSSWizardry

csswizardry.com

CSS - WHY I IGNORED GOOD ADVICE

Normalize but for a brand

Element only style with one exception

Very few media queries

Zero external dependencies!!!!

CSS - WHY?

Backend is not an unmovable block

Low bandwidth version of the site

Zero external dependencies!!!

We wrote a plug to explicitly call this.

SOON TM

JAVASCRIPT

Try breaking up your app from slow moving to fast

Consider Javascript and web sockets as an appreciate experience rather than the only.

It depends. ㄟ_(\ツ)_/

THE OLD WAYS

This is just good fault tolerant architecture



THANK YOU!!!

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THANK YOU!!!

IWANTMYNAME IS HIRING



polymet.is